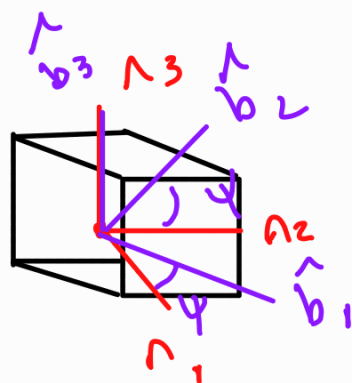


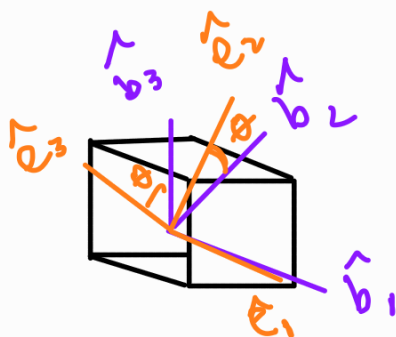
Problem 1

(a) Derive the rotation matrix C for a rigid body in terms of the 3-1-3 Euler angles. These rotate the object first around the $\hat{n}_3$ -axis by an angle ψ , then around the new $\hat{b}_1$ -axis by an angle θ , and finally around the new $\hat{b}_3$ axis by an angle ϕ .

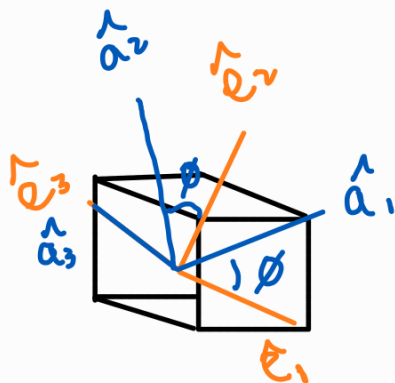
(b) Next, derive the matrix $B(\vec{\theta})$ that defines the Kinematic Differential Equation $\dot{\vec{\theta}} = B(\vec{\theta})\vec{\omega}$, where $\vec{\theta}$ is the vector $[\psi, \theta, \phi]$.



$$R(\psi) = \begin{bmatrix} c\psi & s\psi & 0 \\ -s\psi & c\psi & 0 \\ 0 & 0 & 1 \end{bmatrix}$$



$$R(\theta) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & c\theta & s\theta \\ 0 & -s\theta & c\theta \end{bmatrix}$$



$$R(\phi) = \begin{bmatrix} c\phi & s\phi & 0 \\ -s\phi & c\phi & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$C: R(\phi) \cdot R(\theta) \cdot R(\psi)$$

$$\begin{bmatrix} c\psi & s\psi & 0 \\ -s\psi c\theta & c\psi c\theta & s\theta \\ s\psi s\theta & -c\psi s\theta & c\theta \end{bmatrix}$$

a)

$$C = \begin{bmatrix} c\psi c\phi - s\psi c\theta s\phi & s\psi c\phi + c\psi c\theta s\phi & s\theta s\phi \\ -c\psi s\phi - s\psi c\theta c\phi & -s\psi s\phi + c\psi c\theta c\phi & s\theta c\phi \\ s\psi s\theta & -c\psi s\theta & c\theta \end{bmatrix}$$

b)

$$\vec{\omega}_B = \begin{bmatrix} 0 \\ 0 \\ \dot{\phi} \end{bmatrix} \dot{\phi} + R(\phi) \begin{bmatrix} \dot{\theta} \\ 0 \\ 0 \end{bmatrix} \dot{\theta} + R(\phi) R(\theta) \begin{bmatrix} 0 \\ 0 \\ \dot{\psi} \end{bmatrix} \dot{\psi}$$

$\downarrow$
 $\begin{bmatrix} c\phi & s\phi & 0 \\ -s\phi & c\phi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} c\phi & s\phi c\theta & s\phi s\theta \\ -s\phi & c\phi c\theta & c\phi s\theta \\ 0 & -s\theta & c\theta \end{bmatrix} \begin{bmatrix} \dot{\psi} \\ \dot{\theta} \\ \dot{\phi} \end{bmatrix}$

B^{-1}

$$B = \begin{bmatrix} \frac{s\phi}{s\theta} & \frac{c\phi}{s\theta} & 0 \\ c\phi & -s\phi & 0 \\ -\frac{c\theta s\phi}{s\theta} & -\frac{c\theta c\phi}{s\theta} & 1 \end{bmatrix}$$

Problem 2

Please model the dynamics of an axisymmetric spinning top, spinning about a point fixed P on the ground. To do so, we will use the 3-1-3 Euler angles that you derived in Problem 1. Because the heavy top is axisymmetric, we can take the moments of inertia with respect to P to be $I_t = I_1 = I_2$ and $I_a = I_3$, and choose the principal axes centered at P as our body-fixed axes $\{\hat{b}\}$. The center of mass is a distance l away from P along the b_3 -axis

(a) Write the kinetic and potential energies of the top to find the Lagrangian, and then derive Lagrange's equations of motion.

(b) Two of the three Euler angles are ignorable. Which ones?

(c) Use the Routhian procedure that we've used in the past to get an equation of motion for the only non-ignorable coordinate. Define the constants $a = B_1/I_1$ and $b = B_2/I_2$, with $B_i = \partial L/\partial \dot{q}_i$. Find the 2nd order equation of motion for the non-ignorable coordinate, and the first-order equations of motion for the other two.

$$T = \frac{1}{2} \vec{\omega}^T I_P \vec{\omega}$$

$$I_P = \begin{bmatrix} I_t & 0 & 0 \\ 0 & I_t & 0 \\ 0 & 0 & I_a \end{bmatrix}$$

$$l/z$$

$$\begin{bmatrix} \sin\theta \cos\theta \dot{\psi} + \cos\theta \dot{\theta} & \cos\theta \sin\theta \dot{\psi} - \sin\theta \dot{\theta} & \cos\theta \dot{\psi} + \dot{\theta} \end{bmatrix}$$

$$\begin{bmatrix} I_t & 0 & 0 \\ 0 & I_t & 0 \\ 0 & 0 & I_a \end{bmatrix}$$

$$\begin{bmatrix} \sin\theta \cos\theta \dot{\psi} + \cos\theta \dot{\theta} \\ \cos\theta \sin\theta \dot{\psi} - \sin\theta \dot{\theta} \\ \cos\theta \dot{\psi} + \dot{\theta} \end{bmatrix}$$

$$= \frac{1}{2} (I_t \omega_1^2 + I_t \omega_2^2 + I_a \omega_3^2)$$

$$T = \frac{1}{2} I_t (\omega_1^2 + \omega_2^2) + \frac{1}{2} I_a \omega_3^2$$

$$\omega_1^2 =$$

$$(\sin \theta \cos \theta \dot{\psi} + \cos \theta \dot{\theta}) (\sin \theta \cos \theta \dot{\psi} + \cos \theta \dot{\theta})$$

$$\omega_1^2 = \sin^2 \theta \cos^2 \theta \dot{\psi}^2 + 2 \sin \theta \cos \theta \cos \theta \dot{\psi} \dot{\theta} + \cos^2 \theta \dot{\theta}^2$$

$$\omega_2 = \cos \theta \sin \theta \dot{\psi} - \sin \theta \dot{\theta}$$

$$\omega^2 = \cos^2 \theta \sin^2 \theta \dot{\psi}^2 - 2 \sin \theta \cos \theta \cos \theta \dot{\psi} \dot{\theta} + \sin^2 \theta \dot{\theta}^2$$

$$\omega_1^2 + \omega_2^2 = \omega_3^2 =$$

$$\sin^2 \theta \dot{\psi}^2 + \dot{\theta}^2 \quad \cos \theta \dot{\psi} + \dot{\theta}$$

a)

$$T = \frac{1}{2} I_t (\dot{\phi}^2 + \dot{\theta}^2) + \frac{1}{2} I_a \omega_3^2$$

$$V(\theta) = M g L \cos \theta$$

$$L =$$

$$\frac{1}{2} I_t (\dot{\phi}^2 + \dot{\theta}^2) + \frac{1}{2} I_a \omega_3^2 - M g L \cos \theta$$

ϕ : $\frac{\partial L}{\partial \phi} = 0 \therefore$ ignorable

$$\begin{aligned} \frac{\partial L}{\partial \dot{\phi}} &= I_a (\cos \theta \dot{\phi} + \dot{\phi}) \\ &= I_a \omega_3 = \text{const} \end{aligned}$$

$$\psi: \frac{\partial L}{\partial \psi} = 0$$

$$\frac{\partial L}{\partial \dot{\psi}} = I_t s^2 \theta \dot{\psi} + I_a \omega_3 c \theta = \text{const}$$

$$\theta: \frac{\partial L}{\partial \dot{\theta}} = I_t \dot{\theta} \xrightarrow{\text{d/dt}} I_t \ddot{\theta}$$

$$\frac{\partial L}{\partial \theta} = \frac{1}{2} I_t s 2 \theta \dot{\psi}^2 - I_a \omega_3 s \theta \dot{\psi} + m g L s \theta$$

$$\ddot{\theta} = \frac{1}{2} s 2 \theta \dot{\psi}^2 - \frac{I_a}{I_t} \omega_3 s \theta \dot{\psi} + \frac{m}{I_t} g L s \theta$$

b) ψ and θ are ignorable

$$c) \quad a = \frac{B\psi}{I_t} \quad b = \frac{B\phi}{I_t} = \frac{I_a}{I_t} \omega_3$$

$$a = s^2 \theta \dot{\psi} + b \cos \theta$$

$$\dot{\psi} = \frac{a - b \cos \theta}{s^2 \theta}$$

$$\dot{\phi} = \frac{I_t b}{I_a} - \frac{a \cos \theta - b \cos^2 \theta}{s^2 \theta}$$

$$\ddot{\theta} = \frac{1}{2} s^2 \theta \dot{\psi}^2 - \frac{I_a}{I_t} \omega_3 s \theta \dot{\psi} + \frac{M}{I_t} g l s \theta$$

$$s \theta \cos \theta \cdot \frac{(a - b \cos \theta)^2}{s^4 \theta}$$

$$\ddot{\theta} = \frac{(\cos \theta (a - b \cos \theta))^2}{s^3 \theta} - \frac{b (a - b \cos \theta)}{s \theta} + \frac{M}{I_t} g l s \theta$$